MIL-S-19819C(OS)
25 January 1974
Superseding
MIL-S-19819B(OS)
14 June 1967

MILITARY SPECIFICATION

SWITCH, FIRING, MARK 8 MODS 0 AND 1 (INERTIA)

(For Exploder Mechanism Mark 19 Series)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification states the requirements for the procurement of Firing Switches Mk 8 Mod 0 (LD 291401) and Mk 8 Mod 1 (LD 480486), which are inertia-operated electric switches having normally open contacts. The switches function on target impact to complete a firing circuit.

1.2 Classification. The firing switch described in this specification shall be of one class and one grade but of the following types (see 3.1):

Type I - Firing Switch Mk 8 Mod 0

Type II - Firing Switch Mk 8 Mod 1

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood

FSC 1355

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PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Fiber

Military

MIL-P-116	Preservation, Methods of
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for delivery of
MIL-T-18404	Torpedoes, Environmental Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

LD 291401	Firing Switch Mark 8 Mod 0 (Inertia) and all parts and subassemblies listed thereon
LD 480486	Firing Switch Mark 8 Mod 1 (Inertia) and all parts and subassemblies listed thereon
Dwg 1277873	Firing Switch Subassembly
Dwg 1277875	Weight Assembly
Dwg 1472663	Switch Base Assembly
Dwg 1472712	Switch Base Assembly

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PUBLICATIONS

Naval Ordnance Systems Command

OR-11	Development of Packaging, Handling, Storage and Transportation Systems for Weapons
OS 6341	Miscellaneous General Design Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Firing Switches Mk 8 Mod 0 and Mk 8 Mod 1 shall be inertia-operated, ball-type electric switches with normally open contacts. The switches shall consist of a ball weight assembly and terminal assembly encased in a hermetically sealed metallic shell. One contact pin shall be connected internally to the ball weight, and the other contact pin shall be grounded to the shell. The contact pins shall plug into an electrical receptacle at the base of the cavity where the switch is installed. The Mod O and Mod 1 switches have identical working parts and differ only in one respect: The Mod 0 has an external plastic housing which encases the metallic shell; the Mod 1 has no housing external to the shell.

3.2 General requirements

3.2.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the firing switch shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract..

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the firing switch shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 291401 or LD 480486.

3.2.3 Finishes, coating, construction, and dimensions. Finishes, coating, construction, and dimensions for the firing switch shall be strictly in

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accordance with the applicable drawings, specifications, and publications listed in LD 291401 or LD 480486.

3.2.4 Moisture, fungus, and corrosion. The firing switch shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341 and in accordance with detail requirements specified in the drawings and specifications listed in LD 291401 or LD 480486.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 291401 or LD 480486.

3.2.6 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Definitions. The environmental requirements of this specification are defined in MIL-T-18404.

3.4 Performance requirements and product characteristics. The firing switch shall meet the following performance requirements and product characteristics:

3.4.1 Ball staking. After the ball is staked in position during the assembly of the weight base, the ball shall resist, without perceptible rotation, a 2.5 inch-pound torque applied in either direction about the axis of the shaft.

3.4.2 Pigtail wire securement. The pigtail wire, as secured in the weight base by the bushing, shall withstand without loosening or separating from the base a 1-pound pull applied along the axis of the weight base.

3.4.3 Airtightness. The firing switch subassembly, Drawing 1277873, shall not leak air when the external air pressure is reduced to 15 inches of mercury.

3.4.4 Sensitivity. When the firing switch is accelerated in such a direction that the ball is urged along the main axis toward the nose cap, the switch shall not close with an applied acceleration up to 5.0 gravity units, shall close instantaneously before the applied acceleration reaches 8.0 gravity units, and shall reach a state of steady closure before the applied acceleration reaches 8.5 gravity units. Instantaneous closure shall be considered obtained when a pip is observed on the oscilloscope during testing as specified in 4.4.4.4.3. Steady closure shall be considered obtained when a straight line (with possible hash due to equipment circuitry) is observed on the same oscilloscope screen. When the switch is accelerated in a direction perpendicular to the main axis, it shall not close with an applied acceleration up to 3.5 gravity units, shall close instantaneously

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before the applied acceleration reaches 6.0 gravity units, and shall reach a state of steady closure before the applied acceleration reaches 6.5 gravity units,

3.4.5 Contact resistance. The contact resistance of the switch during steady closure shall be not more than 0.5 ohm, as measured between the two terminals of the switch.

3.4.6 Dielectric strength. When the firing is in the normally open condition, the insulation resistance, as measured between the terminals of the switch, shall be not less than 10 megohms under an applied potential of 500 volts direct current (dc).

3.4.7 Operating life. The firing switch shall have an operating life of not less than 100 complete cycles as defined in 4.3.5.1.

3.4.8 Pressure, external hydrostatic (Type I only). The Type I firing switch assembly shall withstand a pressure of 750 psig, applied to the outer face of the firing switch, without breaking, deforming, or showing evidence of leakage through the housing.

3.5 Environmental requirements. The firing switch shall meet the other applicable requirements of this specification after or while being subjected to environmental conditions defined below, when tested as specified in Section 4.

3.5.1 Vibration

3.5.1.1 Vibration; ready nonoperating or ready condition. The firing switch, less pack or packaging and rigidly self-mounted on a vibration device as specified in 4.3.1.1, shall be capable of withstanding vibrations of 3g's over the frequency range of 10 to 100 Hz in each of three mutually perpendicular planes for time periods and temperatures as specified in 4.3.1.1. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.5.1.2 Vibration: operating condition. The firing switch, self-mounted on a vibration device as specified in 4.3.1.2, shall be capable of withstanding vibrations of 1g over the frequency range of 20 to 150 Hz in each of three mutually perpendicular directions for time periods and temperatures as specified in 4.3.1.2. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

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3.5.1.3 Vibration; transportation. The firing switch, after being subjected to the tests specified in 4.3.1.3, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.2 Shock

3.5.2.1 Shock; ready condition. The firing switch, self-mounted on a shock testing device as specified in 4.3.2.1, shall be capable of withstanding a shock of 150g peak magnitude, of not less than 8 ms pulse duration, and approximating a one-half sine wave pulse. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.5.2.2 Shock; transportation, storage, and handling. The firing switch, after being subjected to the test specified in 4.3.2.2, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.3 Acceleration. The firing switch, mounted securely on an acceleration device as specified in 4.3.3, shall be capable of withstanding an acceleration of 15g's applied for not less than 0.2 second. Thereafter, the switch shall show no damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.5.4 Temperature and humidity

3.5.4.1 Transportation and storage condition. The firing switch, packaged and packed as defined in Section 5, shall be capable of withstanding transportation and storage temperatures and humidities between minus 65 degrees F and plus 160 degrees F for a period of 28 days, as specified in 4.3.4.1. Thereafter, the switch, less pack or packaging, shall show no corrosion or damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.5.4.2 High and low temperature operation. The firing switch, less pack and packaging and mounted as specified in 4.3.4.2, shall be capable of withstanding temperatures over the range of minus 40 degrees F to plus 130 degrees F, as specified in 4.3.4.2. At the conclusion of the test, and while still at the test temperature, the switch shall show no corrosion or damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.6 Cleanup. Prior to and after final assembly, all components shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter.

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3.7 Workmanship. The firing switch, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification, Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, staking, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection of the firing switches shall be classified as follows:

(a) Preproduction inspection (paragraph 4.3)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.3)

4.2 Sampling

4.2.1 Preproduction sample. A preproduction sample of five firing switches manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests detailed in 4.3 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switches by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.2.2 Periodic production sampling. Periodic production samples shall be randomly selected from units that have satisfactorily passed the quality conformance inspections of 4.4. These samples shall be subjected to the tests of 4.3 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

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Step 1. Upon initiation of production randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue Step 3 for the balance of the contract or until a defective unit is found.

4.3 Preproduction and Periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.3.1 Vibration. The firing switch shall be subjected to vibration testing as specified in 4.3.1.1, 4.3.1.2, and 4.3.1.3.

4.3.1.1 Vibration; ready nonoperating or ready condition. The firing switch, mounted as specified on Drawing 1472663 or 1472712, shall be subjected to the vibration testing of Schedule I in each of three mutually perpendicular axes. Thereafter, the switch shall show no damage and shall meet all performance requirements and product characteristics of this specification.

Schedule I

Frequency Range	Amplitude	Minimum Test Duration		
		Room Temp.	-20°F	+130°F
10-30 Hz	0.060 ± 0.006 in.	60 Min.	15 Min.	15 Min.
31-60 Hz	3g ± 0.3g	60 Min.	15 Min.	15 Min.
61-100 Hz	3g ± 0.3g	60 Min.	15 Min.	15 Min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate, over a 3- to 5-minute period for the duration of the test.

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As an alternate method, the firing switch shall be vibrated at specific frequencies, differing by not more than five Hz in each of the test axes and covering the entire frequency range and amplitudes of Schedule I. The total test time shall be not less than that of Schedule I and the test times at the specific frequencies shall be equally divided.

4.3.1.2 Vibration, operating condition. The firing switch, mounted as specified on Drawing 1472663 and 1472712, shall be subjected to the vibration testing of Schedule II in each of three mutually perpendicular axes. During vibration testing, the firing switch shall be continuously monitored to ensure that the switch shows no evidence of closure. At the conclusion of vibration testing, the firing switch shall meet all the performance requirements and product characteristics of this specification.

Schedule II

Frequency Range	Amplitude	Minimum Test Duration Room Temperature
20-60 Hz	lg ± 0.1g	15 Min.
61-150 Hz	1g ± 0.1g	15 Min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate, over a 3- to 5-minute period for the duration of the test,

As an alternate method, the firing switch shall be vibrated at specific frequencies, differing by not more than five Hz in each of the test axes and covering the entire frequency range and amplitude of Schedule II. The total test time shall be not less than that of Schedule II and the test times at the specific frequencies shall be equally divided.

4.3.1.3 Vibration, transportation. The firing switch, packaged and packed as specified in Section 5, shall be vibration tested as specified in OR-11.

4.3.2 Shock. The firing switch shall be subjected to shock testing as specified in 4,3.2.1 and 4.3.2.2.

4.3.2.1 Shock; ready condition. The firing switch, mounted as specified on Drawing 1472663 and 1472712, shall be subjected to three shock pulses of not less than 8 ms duration and approximately one-half a sine wave. The pulses shall have a magnitude of 150 plus or minus 15g’s and shall be applied in both directions along each of three mutually perpendicular axes

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(total of 18 shock pulses). Thereafter, the firing switch shall show no damage and shall meet the performance requirements and product characteristics of this specification.

4.3.2.2 Shock; transportation, storage, and handling. The firing switch, packaged and packed as specified in Section 5, shall be subjected to the appropriate shock isolation tests specified in OR-11.

4.3.3 Acceleration. The firing switch, mounted as shown on Drawing 1472712 or 1472663, shall withstand 15 plus or minus 2g's acceleration for a period of not less than 0.2 second in each of three mutually perpendicular axes with inertial loads in both directions. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

4.3.4 Temperature and humidity

4.3.4.1 Transportation and storage conditions. The firing switch, packaged as specified in Section 5, shall be subjected to transportation and storage, temperatures and humidities between minus 65 degrees F and plus 160 degrees F for a period of 28 days. Thereafter, the switch shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

4.3.4.2 High temperature operation. The firing switch, less pack and packaging, shall be placed within a test chamber and the temperature shall be raised to plus 130 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 plus or minus 5 hours. At the end of this test period, and while still at the test temperature, the switch shall meet all the performance requirements and product characteristics of this specification.

4.3.4.3 Low temperature operation. The firing switch, less pack and packaging, shall be placed in a test chamber and the temperature shall be lowered to minus 40 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 plus or minus 5 hours. At the end of this test period, and while still at the test temperature, the switch shall meet all the performance requirements and product characteristics of this specification.

4.3.5 Operating life

4.3.5.1 Definition of one cycle. One cycle of operation for the firing switch shall be defined as the discharging of a 4-microfarad (mfd) capacitor having an initial charge of 150 volts dc through a 5-ohm resistor with one impact applied to the switch.

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4.3.5.2 Operating life test. Mount the firing switch in a test fixture that will impart sufficient shock to trip the switch. Connect the circuit (4.3.5.1) for discharging the 4-mfd capacitor upon switch closure. Subject the firing switch to 100 cycles of operation, rotating the switch approximately 3-1/2 degrees, with respect to the line of impact, prior to each successive cycle. The minimum time between each impact shall be one minute. The switch shall meet the operating life requirements of 3.4.7.

4.3.6 Workmanship test. Secure the assembled switch in a lathe, or like tool, and cut in half at the 0.062 radius spherical dimple. While carefully separating the halves, look for the presence of dirt, loose or excessive solder, flux, and loose or frayed strands of pigtail wire. This test shall be conducted in accordance with MIL-STD-105, Special Inspection Level S-3, AQL 4.0 percent defective.

4.3.7 Functional tests. At the conclusion of the tests specified in 4.3.1.3 and 4.3.2.2 above, functional and other tests, as applicable, shall be conducted on the firing switches to determine freedom from operational malfunction and compliance with all requirements of Section 3.

4.3.8 Waiver of tests. Tests specified in 4.3.1.3 and 4.3.2.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.4 Quality conformance inspection. Each firing switch delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the firing switch, shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature: Room ambient plus 65 degrees Fahrenheit (F) to plus 95 degrees F (plus 18.33 degrees centigrade (C) to plus 35 degrees C).

4.4.1.2 Altitude: Normal ground.

4.4.1.3 Vibration: None.

4.4.1.4 Humidity: Room ambient to 95 percent relative (maximum).

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, where definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

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4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests as set forth in this specification:

4.4.3.1 A suitable torque-indicating instrument capable of measuring torques in the order of 2.5 inch-pounds.

4.4.3.2 A device for exerting a pull of one pound on a wire in a direction along the axis of the wire.

4.4.3.3 An evacuation chamber that allows observation of its interior, together with a vacuum pump and suitable gauge, capable of registering reduced pressures in the order of 15 inches of mercury.

4.4.3.4 A centrifuge disposed to rotate in a horizontal plane and capable of imparting accelerations of up to 10 gravity units. Provision shall be made for mounting the firing switch in a socket with its axis parallel to the plane of rotation while its terminals are pointed toward the pivotal axis, and with its axis perpendicular to the plane of rotation and its base down. In the perpendicular mounting, additional provision shall be made for rotating the switch about its axis in seven other positions at increments of 45 plus or minus 1 degrees. Sliprings and brushes, or some other appropriate arrangement, shall be provided for making electrical contact with the rotating switch.

4.4.3.5 An oscilloscope whose ripple is limited to less than 0.1 percent. The oscilloscope shall have associated with it an electrical circuit containing a 24-volt dc power source, a 250-ohm resistor, a 1.0-ohm precision resistor, a 0.5-ohm resistor, a single-pole three-position calibration switch, a single-pole single-throw switch, and the firing switch socket on the centrifuge. The dc potential across the 1.0-ohm resistor is applied to the vertical deflection plates of the scope.

4.4.3.6 A megohmmeter with an integral source of 500 volts dc and a midscale reading in the order of 10 megohms.

4.4.3.7 A standard electrical power source of 110 volts alternating current (at), 60 Hz, for operation of the vacuum pump, centrifuge motor, oscilloscope, and other equipment.

4.4.4 Test procedure. The quality conformance tests shall be performed in accordance with the following procedures:

4.4.4.1 Ball staking test. This test shall be performed on Weight Assembly 1277875 before it is hermetically sealed inside the shell. With the use of a suitable torque indicating instrument (4.4.3.1), apply a 2.5 inch-pound torque to the ball, first in one direction and then in the opposite direction about the axis of the switch. When applying the torque, the ball shall not be gripped in such a manner as to scratch, mar, or in any way damage the gold plated surface. The ball shall meet the requirement of 3.4.1.

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4.4.4.2 Pigtail wire securement test. This test shall also be performed on Weight Assembly, 1277875 before it is hermetically sealed inside the shell. Apply a pull of one pound using (4.4.3.2) to the pigtail wire along the axis of the weight base. The pigtail wire shall meet the requirements of 3.4.2.

4.4.4.3 Airtightness test. This test shall be performed upon the Firing Switch Subassembly 1277873 before it is finally assembled into the plastic case. Fill a container with sufficient water so that the subassembly can be completely submerged. Place this water bath in the evacuation chamber (4.4.3.3) and reduce the air pressure to at least 15 inches of mercury in order to remove the air which is dissolved in the bath. Bring the pressure back to ambient and submerge the firing switch subassembly in the deaerated water bath. Evacuate the chamber to a pressure of 15 inches of mercury (maximum), and observe the bath and switch for a period of 2 minutes. Leakage from the switch subassembly will be shown by escaping air bubbles. The firing switch subassembly shall show no leakage, to meet the requirements of 3.4.3.

4.4.4.4 Sensitivity and contact resistance test

4.4.4.4.1 Calibration of oscilloscope. Prior to performing the sensitivity and contact resistance test, it is first necessary to calibrate the oscilloscope (4.4.3.5) to present a one-ohm deflection in fractions thereof. Connect the test equipment as shown on the wiring diagram (Figure 1). The resistor cutout switch shall be in the open position during the entire calibration. Throw the calibration switch to position 1, and adjust the sweep so it is deflected five scale divisions below center scale. Then throw the calibration switch to position 3, and adjust the sweep deflection five scale divisions above center scale. Repeat these steps until the sweep remains properly positioned when the calibrate switch is changed from position 1 to position 3 and the centrifuge is operating. The scope is now calibrated so that each scale division represents one-tenth of an ohm, that is, the relationship between scale deflection and resistance will be linear.

4.4.4.4.2 Conversion of revolutions per minute (rpm) to gravity units. Draw up another table or graph of rpm versus gravity units according to the formula:

$$A = 3.4 \times 10^{-4} RN^2$$

Where the following units apply:

A = gravity units of acceleration

R = Radial arm of centrifuge in feet; the radial arm of the centrifuge shall be defined as the perpendicular distance from the axis of

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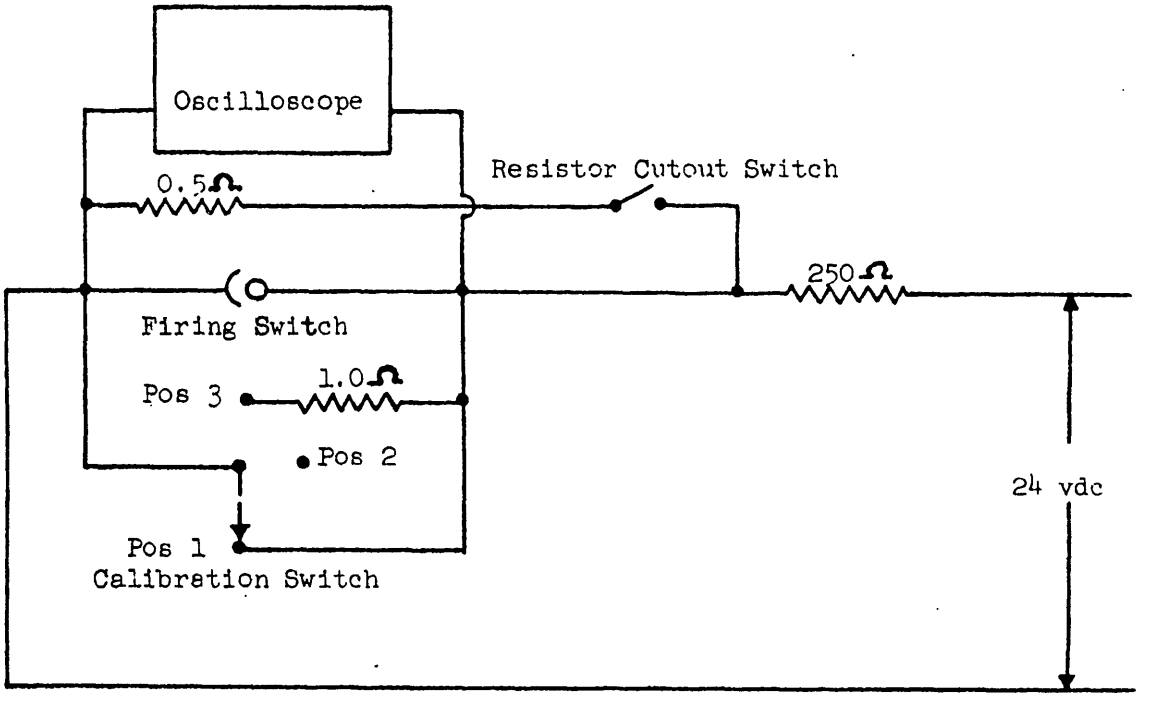


Figure 1. Sensitivity and Contact Resistance Test Wiring Diagram

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rotation to the center of gravity of the moving parts of the switch when the moving parts are at rest.

N = revolutions per minute

If the centrifuge does not register rpm directly, but indicates angular velocity in some arbitrary unit, it will be necessary to obtain a correlation between these units and the rpm.

4.4.4.4.3 Test. Mount the firing switch in the socket on the centrifuge (4.4.3.4) so that its axis is perpendicular to the plane of rotation with its base down. Throw the calibration switch into position 2, close the resistor cutout switch, and start the centrifuge. After attaining an acceleration of 2 gravity units, increase the acceleration thereafter at a rate not exceeding 0.1 gravity unit per second. When a pip is observed on the oscilloscope screen, note the reading on the centrifuge indicator. Convert this reading to gravity units in accordance with 4.4.4.4.2, and record this as the value of instantaneous closure. Slowly increase the acceleration within the specified rate. When a straight line dc presentation is observed on the oscilloscope screen, note the reading on the centrifuge indicator. Convert this reading to gravity units in accordance with 4.4,4.4.2, and record. this as the value for steady closure. Open the resistor cutout switch and record the sweep deflection in ohms resistance for the firing switch during steady closure. If the particular circuit warrants it, short out the firing switch as close to the socket as is physically possible, and obtain a scope deflection for the circuit resistance alone, while the centrifuge is still at the same speed. This resistance shall be subtracted from the value previously obtained to determine true switch contact resistance in the position tested. While perpendicularly mounted, test the firing switch for instantaneous and steady closure acceleration and contact resistance in seven other positions at 45 plus or minus 1 degree increments of rotation about its axis. The switch shall meet the requirements of 3.4.4 (Sensitivity) and 3.4.5 (Contact resistance) in all positions. Check the calibration of the scope deflection before every test. Then mount the switch so that its axis is parallel to the plane of rotation and the switch terminals are toward the pivotal axis of the centrifuge. Conduct one test in this position.

4.4.4.4.4 Dielectric test. Apply the test probes of a megohmmeter (4.4.3.6) to the terminals of the firing switch when it is in the normally open condition. The firing switch shall meet the dielectric strength requirements of 3.4.6.

4.4.4.4.5 External hydrostatic Pressure test (Type I only). Mount the firing switch in a hydrostatic test fixture which will expose only that portion of the switch above the O-ring to the test condition. Apply water pressure to the firing switch housing until a pressure of 750 psig is reached.

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Allow the pressure to remain at 750 psig for a period of 5 minutes. After completing the pressure test, check the outer face of the firing switch for breakage, deformation, or leakage through the housing. None of these conditions shall be indicated, in accordance with 3.4.8.

4.4.4.5 Visual and dimensional examination. Each firing switch shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.6 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine conformance with the requirements of Section 5.

5, PREPARATION FOR DELIVERY

5.1 Packaging and packing

5.1.1 Level A. Unless otherwise specified, the firing switch shall be packaged and packed as an integral component of the exploder mechanism in the containers specified for the exploder mechanism as set forth in the applicable NAVORD drawings.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Not applicable.

5.2 Spare or replacement parts, packaging and packing. When procured as spare or replacement parts, firing switches shall be packaged as follows for delivery to the using activities:

5.2.1 Level A. In addition to the following, Level A packaging shall be in accordance with MIL-E-17555.

5.2.1.1 Cleaning, drying, and preservative application. Not applicable.

5.2.1.2 Unit packaging. Unit packaging of the firing switch shall be in accordance with Method 1A-5 of MIL-P-116. The unit package shall contain one each.

5.2.1.3 Intermediate package. Not applicable.

5.2.2 Level B. Not applicable.

5,2.3 Level C.

5.2.3.1 Cleaning, drying, and preservation application. Not applicable.

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5.2.3.2 Unit packaging. Unit packaging of the firing switch shall be in accordance with Method III of MIL-P-116. The unit package shall contain one each.

5.3 Packing

5.3.1 Level A. In addition to the following, Level A packing shall be in accordance with MIL-E-17555.

5.3.1.1 Exterior containers. The firing switch, packaged as described in 5.2.1.2, shall be packed in a wood box in accordance with PPP-B-621, Class 2 or PPP-B-601, overseas type or in a wirebound box in accordance with PPP-B-585, Class 3 (weight limit not to exceed 150 lbs), or a fiberboard box in accordance with PPP-B-636, Class 2 or PPP-B-591, overseas type (weight not to exceed 70 lbs).

5.3.2 Level B. In addition to the following, Level B packing shall be in accordance with MIL-E-17555.

5.3.2.1 Exterior containers. The firing switch, packaged as described in 5.2.1.2, shall be packed in a wood box in accordance with PPP-B-621, Class 1, or PPP-B-601, Domestic Type, or in a wirebound box in accordance with PPP-B-585, Class 1 (weight limit not to exceed 150 lbs) or in a fiberboard box in accordance with PPP-B-636, Class 1 or PPP-B-591, Domestic type (weight limit not to exceed 70 lbs).

5.2.3 Level C. The firing switch, packaged in accordance with 5.2.3.2, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Levels A, B, or C cushioning. The unit package or pack shall incorporate sufficient cushioning material, bracing or other adequate shock absorbing devices to ensure the firing switch will meet all the requirements of Section 3 after being tested in accordance with 4.3.1.3 (Vibration, transportation) and 4.3.2.2 (Shock; transportation, storage, and handling). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings

5.4.2 Normal markings. In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping

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containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. Firing Switch Mk 8 Mod 0 and Firing Switch Mk 8 Mod 1 are intended for use in the Exploder Mechanism Mk 19 series.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exception to this specification.

(c) Type of firing switch required (see 1.2).

(d) Required level of preservation and packaging.

(e) Required level of packing.

(f) Assigned activity for preproduction and periodic production inspections.

(g) That addition samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier,

6.2.1 The attention of the contracting officer is invited to the quality assurance provisions and options in MIL-P-116 and to 4.3.8 of this specification.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDS
4.3	Preproduction and periodic production inspection	

DIDS (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project No. 1355-N325

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-19819C(OS), Switch, Firing, Mark 8 Mods 0 and 1 (Inertia)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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Standardization Division (502)
Indian Head, MD 20640

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